

Suggested hardware for the transceiver

From Radio Team (Santiago, Andrian, Marcos) **To** Carlos, Michele, Bikash, Guillem

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The radio baseline is 868 MHz with the RFM69HCW. Under our split, the electronics team picks the micro and designs the board. This page is the radio team's **suggestion**, built around every hardware rule in the competition PDF, to save you time. It is not a decision. Please challenge it.

In short

1. **Micro module: Raspberry Pi Pico 2** (RP2350, **not** the Pico 2 W). It costs about \$5, has 4 MB flash to hold the 1 MB file, 520 KB RAM and 3.3 V logic, and it can act as a USB host on any two pins or on its native USB port. It has no radio chip on board, so it is safe under the protocol ban.
2. **Radio module: RFM69HW/HCW-868** (SX1231H), about \$3.5. The radio team's driver for it is already written and tested.
3. **Board: a custom 2-layer carrier PCB** that the Pico 2 and the radio solder onto. It also carries the USB-A host port, the battery charger, the 5 V supply for the stick, ESD protection, the switch, button and LEDs, and a u.FL antenna connector. Before Quick Mode we use a perfboard prototype with the **same modules**.
4. **Power: one protected 18650 Li-ion cell**, charged on board through a charge-only USB-C port. Our whole competition day uses roughly 10–20 % of it.
5. **Estimated cost: about €45–55 per device** with a lab-made PCB, or about €30–40 with an external fab, against the €150 cap. This keeps us competitive in the cost category.

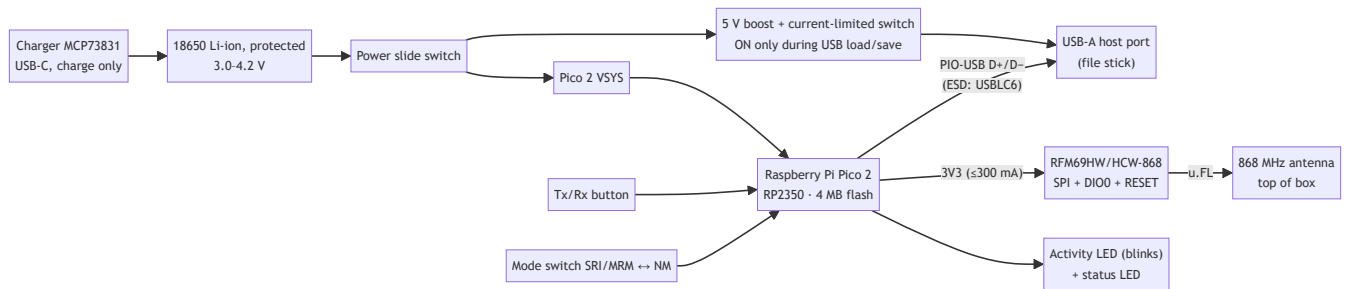
The rules this design answers

RULE (COMPETITION PDF)	WHAT IT FORCES ON THE BOARD
One and only one module including a microprocessor; no single commercial	Pico 2 is the only micro. The radio, charger and boost contain no CPU. No USB-host chips that have their own controller

RULE (COMPETITION PDF)	WHAT IT FORCES ON THE BOARD
module embedding the entire system	(e.g. CH376), which could count as a second micro.
No chip implementing WiFi, Bluetooth, ZigBee, 2G–5G	Banned even if unused: Pico W / Pico 2 W, ESP32 boards, Arduino Nano 33 BLE/IoT, Uno R4 WiFi, Nano ESP32, Raspberry Pi Zero W / 3 / 4 / 5.
Read/write the file from/to a USB memory device: 5 min setup while still wrapped in foil, 3 min to upload afterwards	A USB-A host port on the box wall, plus ≥ 1 MB on-board storage . The sender copies the file in at setup, and the receiver keeps what it got until upload. The stick can be removed during the round.
Fits entirely in the 15 × 15 × 7 cm IKEA 365+; only connectors, switches and buttons may stick out	PCB about 100 × 80 mm or smaller. The antenna sits inside, connected by u.FL so it can be placed high in the box.
Operates autonomously	A battery and a power switch. Nothing plugged in during a round.
At least one Tx/Rx switch or button; a pilot light that blinks in Tx/Rx	Panel pushbutton for Tx/Rx and an activity LED. The box is transparent, so the LED can stay inside.
Extra switches allowed (e.g. network mode), but no switch indicating the position in the network	One optional SRI/MRM ↔ NM mode switch. Nothing that encodes a node number.
Drop test from table height to concrete (before MRM)	Everything soldered (no sockets or loose headers), board on standoffs, battery strapped, antenna glued down.
ESD and RF exposure compliance	TVS/ESD protection on USB and the buttons. RF power is small (~27 mW e.r.p.).
Modules shown at Quick Mode are final	The QM prototype uses exactly the Pico 2 and RFM69 part numbers of the final board.
Unit cost ≤ €150; lab-made PCB counts €15; everything (screws, foam, wires) goes in the BoM	Cheap modules; consider an external fab for the PCB (see the cost table).

Block diagram

The only fast data path is SPI to the radio, at 12.5 kB/s. USB file transfer happens only before and after a round.



Micro module options

Ranked against what the board must do: hold a 1 MB file, act as a USB host, use 3.3 V logic, carry no radio chip, stay cheap, and be ready before Quick Mode.

MODULE	STORAGE FOR 1 MB FILE	USB HOST	PRICE	VERDICT	NOTES
Raspberry Pi Pico 2 (RP2350)	4 MB flash ✓	PIO-USB on 2 GPIOs, or native USB	~\$5	recommended	Native USB stays free for programming and logs. Big community. Castellated edge, so it solders flat (good for the drop test).
Raspberry Pi Pico (RP2040)	2 MB flash, tight	Same as above	~\$4	fallback	Works, but 1 MB of file plus firmware leaves little headroom.
Adafruit Feather RP2040 USB Host	8 MB flash ✓	USB-A on board	~\$15	fastest prototype	Less to build, but costs more, and we would show less electronics design of our own (which is graded).

MODULE	STORAGE FOR 1 MB FILE	USB HOST	PRICE	VERDICT	NOTES
Teensy 4.1	8 MB flash ✓	Host port + library	~\$35	expensive	Easiest USB host, but it hurts the cost category.
STM32F411 "Black Pill"	512 KB ✗	Native OTG	~\$5	needs extra flash	Would need external flash, and USB mass-storage host firmware is heavier.
Arduino Uno / Nano (AVR)	✗	✗	—	no	5 V logic (the radio is 3.3 V only), 2 KB RAM, no USB host.
Raspberry Pi Zero (non-W)	SD card ✓	✓	~\$5–10	no	Linux: slow boot, high current, and SD corruption on power loss or a drop.
Pico W / Pico 2 W, ESP32, Nano 33 BLE/IoT, Uno R4 WiFi, Pi Zero W, Pi 3/4/5	—	—	—	banned	On-board WiFi/BT chip.
Micro + radio combos (Feather RP2040 RFM69, Moteino)	—	—	—	avoid	Risks the "single commercial module embedding the entire system" rule. Would need an Atenea answer first.

Prices are list prices seen online on 7 Oct 2026 (Raspberry Pi, TME, Adafruit). Get real quotes before ordering.

Suggested parts per device

Example part numbers to start from. Electronics owns the final choice, the footprints and the sourcing.

FUNCTION	SUGGESTED PART	WHY	≈ € / DEVICE
Micro module	Raspberry Pi Pico 2 (SC1631), non-W	See above	5.5
Radio module	HopeRF RFM69HW-868S2 or RFM69HCW-868S2	Same SX1231H radio, different pad layout; pick one footprint and keep it	3.5–6
Antenna	u.FL connector + $\lambda/4$ wire (QM), 868 MHz FPC or helical for the A/B test	u.FL lets the antenna sit high, away from the PCB	0.5–5
USB host port	USB-A receptacle, through-hole, panel-facing	Through-hole survives plugging and the drop test	0.5
5 V for the stick	Boost (e.g. TPS61023) + current-limited switch (e.g. TPS2051 / AP2552), enabled by a GPIO	Sticks need 5 V. Powered only during load/save, which saves energy and protects against shorts	2
ESD	USBLC6-2SC6 on D+/D–; TVS on VBUS and the button lines	ESD compliance; the USB port is touched every round	0.6
Battery	1 × protected 18650 Li-ion (~2500 mAh) with solder tabs, or holder + strap	Steel can is robust in the drop test; rechargeable (sustainability report)	6–8
Charger	MCP73831 (or TP4056) + USB-C receptacle, charge only	Charge between rounds without opening the box. Nothing is plugged in during operation	1.2
Controls	Slide power switch, panel Tx/Rx pushbutton, optional mode switch, 2 LEDs	Required by the rules	2.5
Bulk capacitance	47–100 μF + 100 nF at the radio; 100 μF on USB VBUS	Handles ~95 mA radio TX bursts and stick inrush	0.5
PCB	2-layer carrier, $\leq 100 \times 80$ mm	ETSETB lab = €15 in the BoM (rule); an external fab is ~€1 per board but adds 1–2 weeks	1–15

FUNCTION	SUGGESTED PART	WHY	≈ € / DEVICE
Mechanics	M3 nylon standoffs, base plate, cable ties, hot glue, IKEA 365+ box	Drop test; the BoM must list them too	5–7
Total per device (verify with quotes)			≈ 30–55

Power budget

Current draw

Pico 2 running	≈ 30–50 mA	ESTIMATE
RFM69 TX +17 dBm	≈ 95 mA	DATASHEET
RFM69 RX	≈ 16 mA	DATASHEET
USB stick (5 V)	50–100 mA, only during load/save	ESTIMATE
Peak from 3V3	≈ 150 mA, under the Pico's 300 mA limit	

Battery sizing

A sending round averages about 140 mA. Idle with the radio listening is about 50 mA. A full competition day of ~4 hours powered on needs about **250–400 mAh**.

One 2500 mAh 18650 covers that about six times over. A smaller LiPo would also do, but pouch cells are more fragile in the drop test.

For the sustainability report

We can measure watts per round directly on this design. The parts are rechargeable and RoHS. The modules unsolder and can be reused, and the board shape lets you repair rather than replace.

Build stages

STAGE	BOARD	GOAL	TARGET
1 · Bench	Pico 2 + RFM69 on a 2 mm→2.54 mm adapter, breadboard, USB powered	Radio tests T0/T2; USB-stick read/write via PIO-USB	as soon as parts arrive

STAGE	BOARD	GOAL	TARGET
2 · Quick Mode	Perfboard, same module part numbers , external power allowed	One error-free line from a file in < 2 min, first attempt	week 6–7
3 · PCB rev A	Custom 2-layer carrier (KiCad), boxed, battery	Autonomous, drop-tested, range tests T4/T5	week 8–9
4 · PCB rev B	Fixes from rev A	Pretest (4 Dec) fully packaged	week 10–11

Risks and how we cover them

RISK	MITIGATION
PIO-USB host fails with some USB sticks	Test 3–4 different sticks in stage 1. The RP2350's native USB can also be a host, so a second path exists on the same chip.
The foil-wrapped setup makes the USB port hard to reach	Put the port on a side wall. Ask on Atenea whether the foil may be opened at the port during the 5-min setup.
Brown-out during radio TX	Bulk capacitance at the radio; check the battery at low charge (3.3 V) in stage 3.
Drop test breaks a joint or ejects the battery	No sockets; standoffs; strapped cell; drop rev A on purpose in week 9.
The "attacks" interrupt the link mid-round	Watchdog and brown-out reset; firmware resumes from the last acknowledged line (protocols team).

What we need from electronics

1. Agree or counter-propose the micro by **Fri 16 Oct**, so we can order together with the radio modules.
2. Pick the RFM69 footprint (HW-S2 or HCW) and the PCB route (lab at €15, or external fab).
3. Pin map for SPI, NSS, RESET, DIO0, PIO-USB D+/D–, button, LEDs and mode switch. We will write the radio's hardware layer against it.

4. Book Prof. Chávez or Prof. Jiménez to review this block diagram and the lab's PCB capabilities.

Radio details (interface pins, timing, test plan): Team C Radio Brief v0.2. Rules quoted from *MTP F26 – Program Competition Rules – Network Upgrade III + Sustainability*.

Sources checked 7 Oct 2026: [Pico-PIO-USB](#) (USB host/device via PIO, RP2040 and RP2350) · [Pico 2 specs](#) · [TME RFM69HW-868S2](#).